

|                                                             | S1<br>sense | S2<br>geom.<br>est. | S3<br>channel<br>est. | S4<br>optimise | S5<br>control<br>tx | S6<br>electronic<br>update | S7<br>mechanical<br>morphing | S8<br>settling | S9<br>calibration | S10<br>stabilised<br>RF |
|-------------------------------------------------------------|-------------|---------------------|-----------------------|----------------|---------------------|----------------------------|------------------------------|----------------|-------------------|-------------------------|
| A1 theoretical FIM<br>[1], [2], [3], [15]                   |             |                     | S·T                   | S              |                     | A                          | A                            |                |                   | A                       |
| A3 flexible programmable aperture<br>[6] (geometry imposed) | Q·T         | Q·(T)               |                       | Q·T            | Q·(T)               | Q·T                        |                              |                | D                 | Q·T*                    |
| A5 rigid reconfigurable RIS<br>[4], [5]                     |             |                     |                       |                | Q·(T)               | Q·T                        |                              |                |                   | Q                       |
| A4 self-morphing mechanical<br>[7], [8] (no RF layer)       | Q·T         | Q·T                 |                       | Q·T            | Q·T                 |                            | Q·T                          | Q·T            |                   |                         |
| A7 flexible active array<br>[20]                            | D           | D                   |                       | D              | D                   | D                          |                              |                | D                 | Q                       |

·T the duration of the stage was measured, with identifiable start and end events

·(T) a duration is reported but incompletely defined, or not resolved from an adjacent stage

\* geometry held static throughout the timed interval

across the reviewed set, these four stages are timed only on platforms with no RF layer (S7, S8), not timed at all (S9), or timed only with the geometry held static (S10)

■ Q quantitatively measured    ■ A / S assumed or simulated  
 ■ D demonstrated, no quantity    ■ absent / not applicable